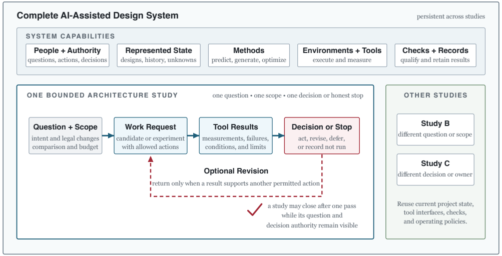
SYNTHESIS LECTURE

Architecture 2.0

Principles for AI-Assisted System and Chip Design

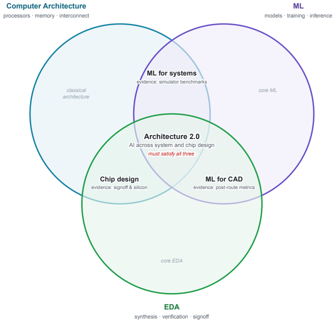


Vijay Janapa Reddi

arch2.mlsysbook.ai

Three fields, one architecture decision

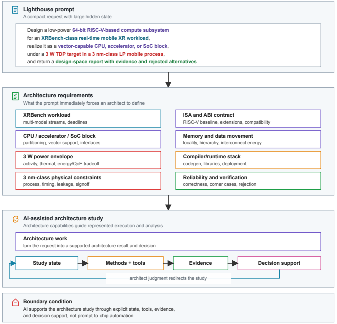
The useful intersection is not “AI for chips.” It is a supported architecture decision.



- Architecture supplies the tradeoff and system boundary.
- ML supplies possible prediction, generation, and optimization methods.
- EDA supplies executable constraints and physical observations.

The one-sentence trap

A compact product request hides a full-stack architecture program.



Source: Architecture 2.0, Chapter 1; XRBench and ILLIXR ground the workload class.

Lighthouse target

- low-power, 64-bit RISC-V compute subsystem
- real-time mobile extended-reality (XR) workloads
- software, RTL, physical, verification, and product constraints

Four different success claims

Output, architecture value, AI contribution, and moonshot completion are not interchangeable.

Candidate
A legal artifact exists

→

Architecture result
The comparison supports a change

→

AI contribution
The complete approach helped

→

Moonshot
The capability holds across the target

Each arrow needs a separate test.

The Architecture 2.0 bet

Improve the architecture result, reduce the total cost of reaching a comparable result, or both.

Architecture quality

- performance, power, area
- reliability, security
- software and product fit

Total cost

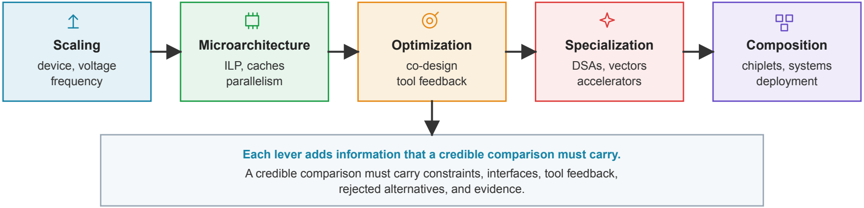
- model and tool calls
- queue and license time
- human setup, review, repair

Failure behavior

- invalid candidates
- missed constraints
- unsafe or uncontrolled actions

Every efficiency lever adds obligations

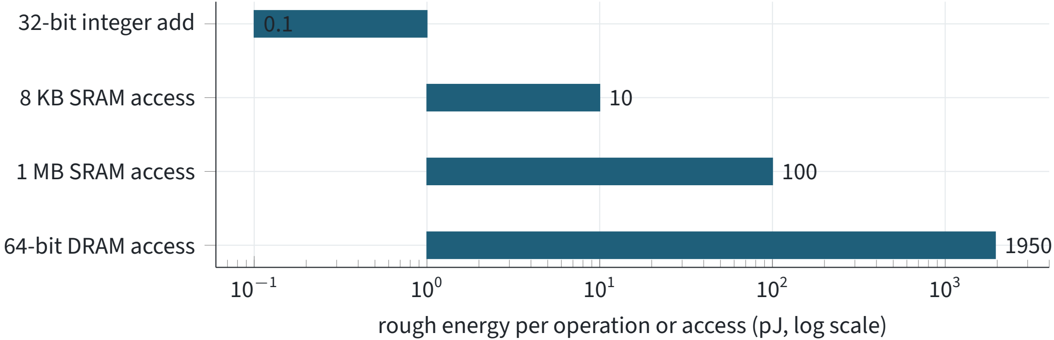
Scaling gave way to specialization, composition, and cross-layer work.



Source: Dennard scaling, Amdahl's law, dark silicon, chiplets, and warehouse-scale computing are established architecture results summarized in Chapter 2.

Data movement can dwarf arithmetic

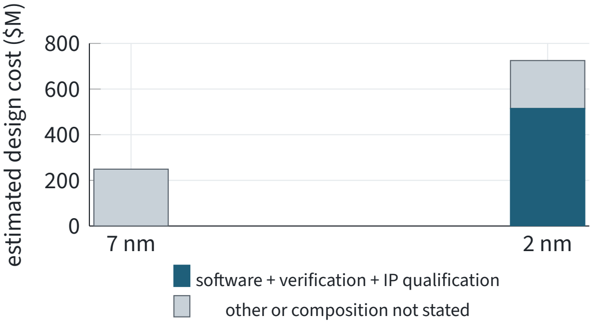
The system boundary decides whether a local compute gain matters.



Source: Horowitz, ISSCC 2014 rough 45 nm energy anchors. DRAM is plotted at the midpoint of the reported 1300–2600 pJ range; these are order-of-magnitude anchors, not current-node predictions.

Cost has moved beyond drawing the RTL

At advanced nodes, software, verification, and IP qualification can dominate estimated design cost.



71%

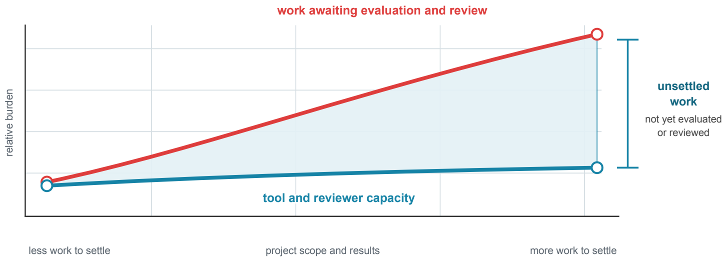
of the $725M 2 nm estimate was attributed to the combined software, verification, and IP-qualification categories.

The 7 nm composition was not published, so the chart does not invent it.

Source: Arm Holdings, Form 424B4 (2023), reporting July 2022 IBS estimates: $249M at 7 nm and $725M at 2 nm.

Production can outpace examination

Faster candidate creation is useful only if evaluation and review can keep up.



The bottleneck is evaluation and review capacity, not idea generation.

This is one possible limiting pattern; the next slide diagnoses other cases.

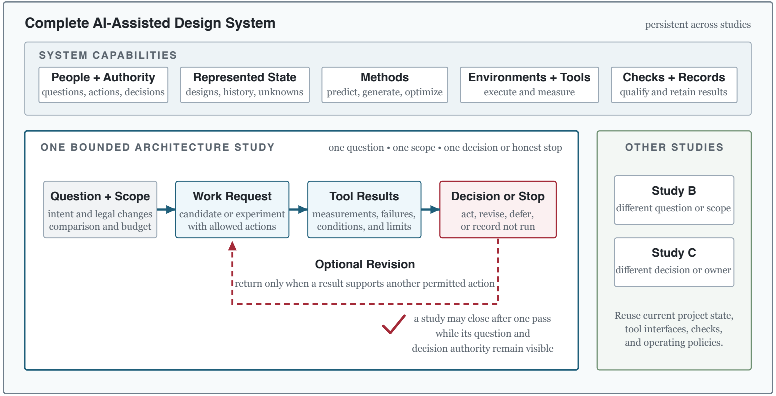
Diagnose the limiting work first

Different delays call for different interventions.

Missing information	retrieve, reconcile, or collect the state
Invalid or unhelpful choices	constrain or change the candidate space
Expensive evaluation	predict, screen, or allocate stronger checks
Overloaded review	change capacity, routing, or review preparation
Unclear decision	name the action, owner, and evidence threshold

“Use AI” is not a diagnosis.

The object being engineered is the complete system



System, study, and loop are different

The loop is a possible behavior inside a study, not the whole thesis.

Complete system

- persistent people, methods, tools, state, checks
- supports many studies

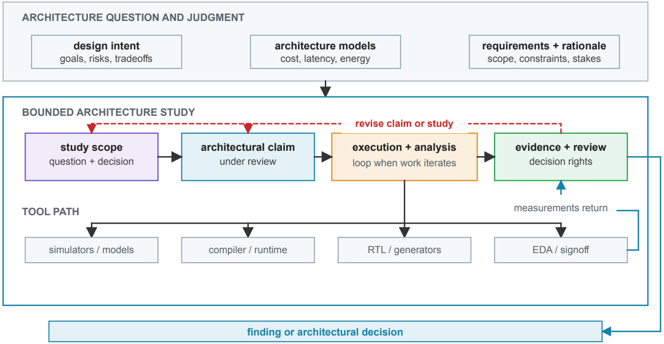
Bounded study

- one question and scope
- one decision or honest stop

Optional loop

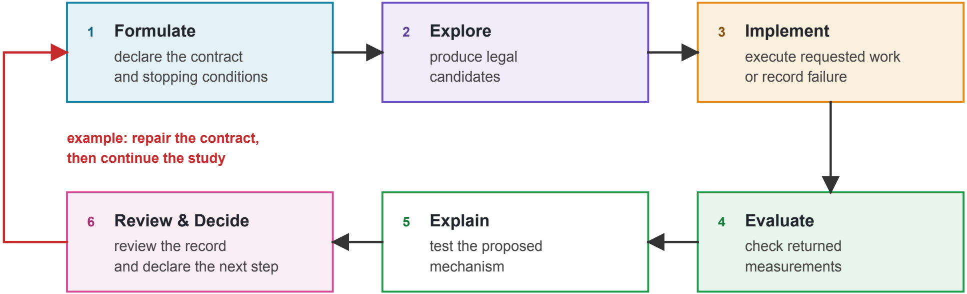
- another permitted action
- only when the result supports it

A bounded study connects question to decision



Six distinct kinds of work

The stages are separated because they answer different questions and fail differently.



Repair the part that failed

Bad question reformulate	**Weak comparison** change alternatives or budget	**Broken execution** repair the environment
Inadequate check strengthen measurement	**Failed explanation** narrow or retest mechanism	**Unresolved decision** defer, reject, or assign work

Iteration should have a technical reason and a destination.

One record holds the comparison together

Another architect should be able to reconstruct what changed, what ran, what failed, and what the result supports.

Question + permitted changes	Comparator + budget	Environment + tool conditions
Accepted and rejected candidates	Measurements, failures, and limits	Supported result + accountable action

The record connects the technical argument; it is not the argument itself.

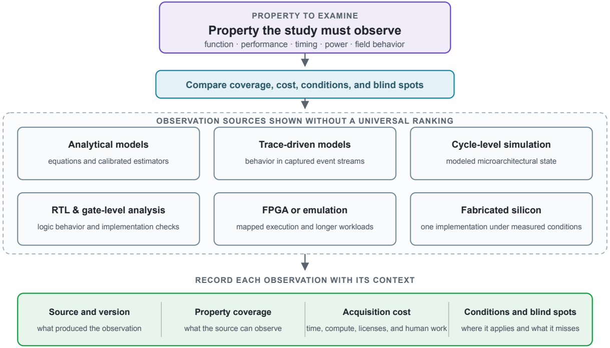
The project knows three different things

General knowledge	mechanisms, specifications, papers, slow-moving relationships
Current project state	exact design, workload, software, constraints, owners
Project history	runs, failures, rejections, adoption, changed conditions

A model's general knowledge is not live project state.

Observation sources are not interchangeable

Choose the source by the property the study must observe, not by a single fidelity ranking.



An architecture sample can be expensive

Data acquisition is part of the architecture method, not background plumbing.

994
legal microarchitectures

→

6–14+ h
per VLSI-flow evaluation

→

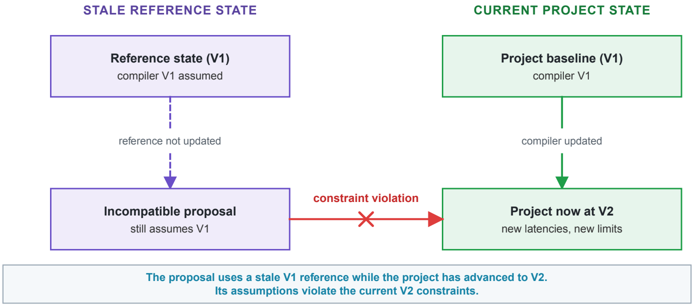
$\geq$5,964 h
aggregate evaluation time
at the reported low end

Ten analytical calls, ten cycle simulations, and ten physical-design runs are not ten equivalent samples.

Source: BOOM-Explorer: Bai et al., ICCAD 2021. The final value is 994 $\times$ 6 h: a serial aggregate, not elapsed wall time.

Representation determines reachable designs

A legal proposal can still be invalid when project state and tool-facing artifacts disagree.



Role comes before method

Do not collapse an engineering job, a method family, and an implementation into "the AI."

1. Engineering role	propose, estimate, select, check, operate, interpret
2. Need for an added method	direct tools or existing analysis may already suffice
3. Method family	generation, prediction, optimization, or a combination
4. Implementation	classical, learned, language-model based, or hybrid

Use the smallest sufficient approach

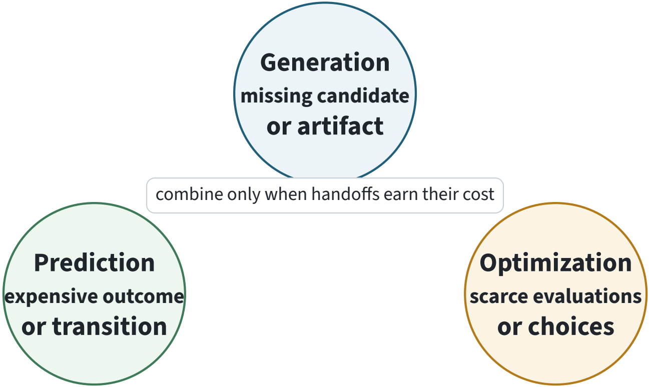
Choose the least costly approach that resolves the named bottleneck.

Direct analysis closed form, roofline	**Enumeration** when the space fits	**Established tools** compiler, simulator, EDA	**Added method** generator, predictor, or optimizer

An added generator, predictor, or optimizer is optional when direct analysis or tools suffice.

Three method families

Start from what is missing, not from a fixed prediction–generation–optimization sequence.



Generation expands; verification narrows

Attempted, executable, test-passing, and useful are different conditions.

100 generated kernels

83 compile

73 execute

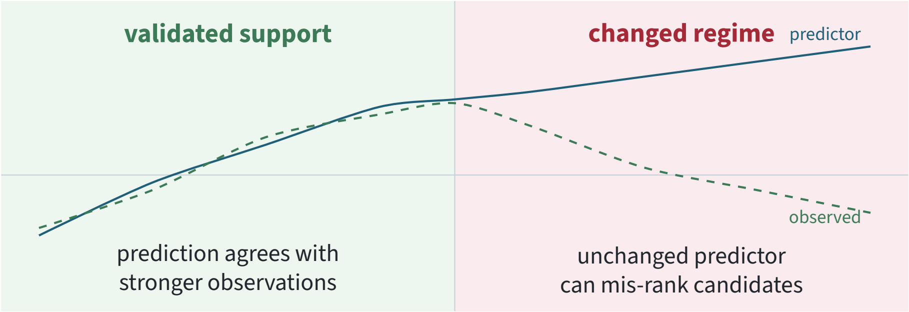
50 match KernelBench's reference outputs on five test inputs

Passing five tests does not establish correctness outside those inputs.

Source: KernelBench, Ouyang et al. (2025), separate 100-kernel error-analysis sample. Per-model one-shot results in the paper show the same attempted → correct → correct-and-faster narrowing.

Prediction trades cost for a support assumption

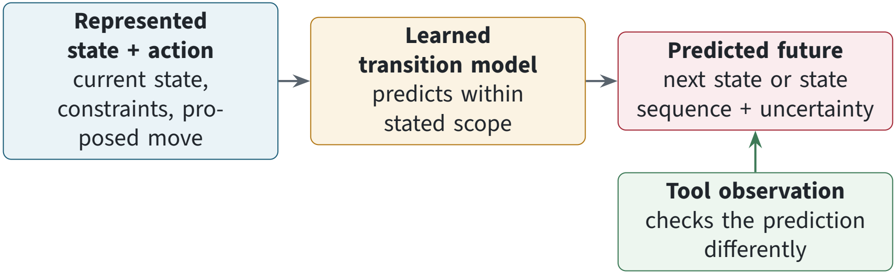
A cheap predictor helps only where its error is acceptable for the decision it screens.



Source: Constructed teaching illustration; Chapter 5 develops calibration and support limits.

A world model is a learned transition model

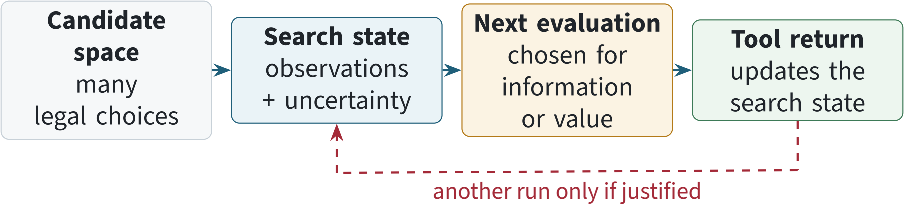
It predicts a sequence of future states from a represented state and proposed actions. Here it supports prediction or planning; it is not the live project record or a verification tool.



The observation can test model support; it does not make the model authoritative.

Optimization can allocate the next evaluation

Optimization selects or improves candidates under an objective, constraints, and a stopping rule. For costly black-box objectives, it must also decide what to measure next.



Method handoffs must earn their cost

This is one possible combination; no fixed order is implied.

Generate a legal candidate → **Predict** screen outcome or cost → **Optimize** choose the next run → **Direct evaluation** invoke the decisive tool

Handoffs carry assumptions about representation, support, and budget. Tool returns still need qualification.

Justified

- each stage removes more cost than it adds
- rejected candidates and failures stay visible

Not justified

- more components without a named bottleneck
- a cheap proxy replaces the decisive check

An environment is more than a simulator

An environment defines how a design request becomes a recorded tool outcome.

Inputs
question + workload
permitted actions
constraints + metrics

→

Environment
check request
invoke the tool
parse status
record cost + provenance

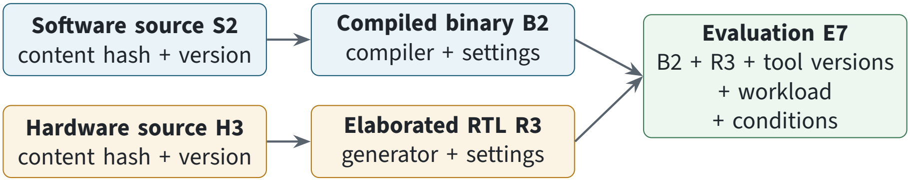
→

Recorded outcome
invalid request
execution failure
incomplete or complete return
raw output + run record

Define permitted actions, fixed context, expected outputs, cost, lineage, and status before execution.

Identity must survive every tool

A successful command is not enough to reconstruct a result.



Artifact identity = content + production lineage.
Evaluation identity = exact artifacts + tool versions + workload + conditions.

A return is not yet a measurement

A value becomes a measurement only when its identity, conditions, status, and limits are bound to it.

1. **Tool return** status, logs, artifacts
2. **Bind identity** candidate, property, unit
3. **Bind conditions** tool + version, config, artifact status
4. **State limits** variation, error, or known bound
5. **Compare + interpret** baseline, meaning, action

If qualification fails, retain an execution return, not a measurement for comparison.

Formal and simulation answer different questions

Formal

- proves a stated property or finds a counterexample
- under a model and assumptions
- may be inconclusive or time out

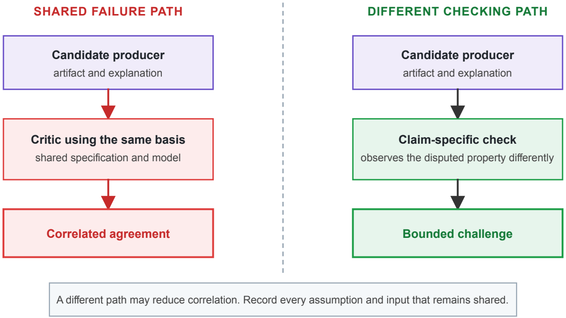
Simulation

- observes exercised behavior
- under selected stimuli and configuration
- cannot exhaust a modern design

Neither check repairs an incomplete requirement or system boundary.

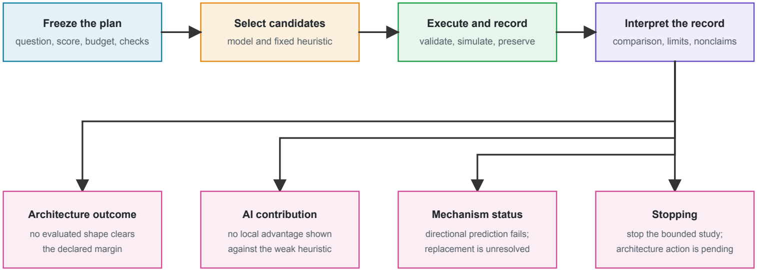
Another model is not an independent check

Independence comes from a materially different observation path, not another fluent answer.



One complete study must resolve four judgments

Architecture outcome, AI contribution, mechanism, and action remain separate.



Equal simulator budgets produced a tie

Frozen local score = total modeled cycles ÷ average layer utilization; lower is better. It is a study-local ordering rule, not a physical measurement.

Model arm	4 simulator calls	score 33,782	32×32 and 16×64 tied
Fixed heuristic	4 simulator calls	score 33,782	32×32 and 16×64 tied

Outcome: no evaluated nonbaseline cleared the margin.
Recommendation: retain 32×32; architecture action pending.

Post-run check: 16×64 had twice the baseline's modeled DRAM-word writes, omitted from the frozen score.

Source: Retained SCALE-Sim 3.0.0 study in Chapter 8. Cycle-level simulator; local lower-is-better score; no silicon measurements.

The failed explanation is also a result

In this general matrix multiplication (GEMM) contrast, the local comparison survives even though the proposed explanation fails.

Setup

In a weight-stationary dataflow, weights stay in the array. K maps to array height, N to array width, and M streams through.

16×64 array
height 16, width 64

64×16 array
height 64, width 16

Matched workload contrast: swap M and N; keep K unchanged.

Prediction

Transposing M and N would reverse the winner.

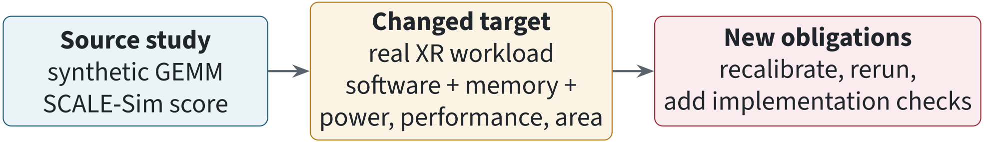
Observed

16×64 still wins the matched contrast.

Keep the outcome. Reject the mechanism claim.

A new decision reopens the claim

Study structure may transfer; a local conclusion does not transfer automatically.



Workload, software, hardware, tools, process assumptions, and decision scope are part of the result.

Evaluate matched complete systems

A model score cannot establish that the complete AI-assisted approach improved architecture work.

Same decision, starting information, evaluator-owned tools and checks, cost window, and stopping rule

AI-assisted complete system

- internal models, methods, and tools
- operator expertise and intervention
- failures and all resource costs retained

Strong comparison system

- established heuristic or human process
- practitioner expertise and intervention
- failures and all resource costs retained

Compare paired outcome–cost results across declared budgets and repeats. Report uncertainty and AI contribution separately.

Metrics people forget to count

The result, the work that produced it, and its decision quality need separate measurements.

Architecture	performance, power, area, reliability, software and product constraints
Work performed	candidates, model calls, tokens, tool calls by type, unique simulator or implementation runs
Resources + people	wall/queue time, compute, energy, licenses, data preparation, training, setup, review, repair
Experimental validity	task/workload coverage, paired repeats/seeds, dispersion, leakage/contamination checks
Evidence + decision	qualification, uncertainty, check coverage, supported or unresolved status, reopening
Failures + control	invalid actions, retries, timeouts, recovery, budget violations, unsafe changes

Ten tool calls do not have one common cost.

Stress-test failures; red-team attacks

Fault injection probes resilience. Red teaming predeclares the asset, adversary, access, attack goal, success criteria, containment, and recovery.

Artifact attack
hidden logic, backdoors, unsafe interfaces, physical side effects

Evaluator attack
weakened tests, proxy gaming, omitted conditions, contaminated data

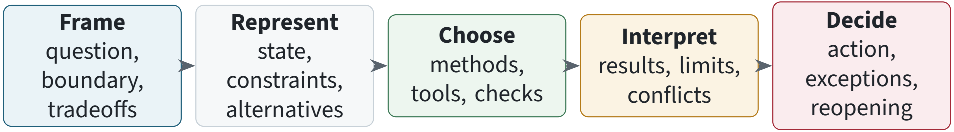
Workflow faults
silent retries, hidden failures, uncontrolled tool or model spending

Authority attack
changed requirements, unauthorized waivers, recommendation treated as approval

Source: Chapter 10. Check final artifacts and downstream tool stages, not only source RTL; record attack success, detection, containment, recovery cost, and residual risk.

What remains architectural

The architect's role is a positive technical contribution, not whatever automation has not absorbed.



Specialists supply capabilities. The architect integrates them into a supported decision.

Three separations to remember

A candidate is not an architecture result.

A tool return is not yet a measurement.

A favorable assessment is not permission to commit.

The responsible authority must commit, refuse, or keep the decision unresolved.
Residual risk, containment, and reopening conditions remain part of the record.